

NON-SMOOTH MECHANICS FOR FOUNDATION POLICIES

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ABSTRACT

Robot foundation policies are usually trained and evaluated as smooth input-output maps, yet contact-rich manipulation is full of non-smooth events: impacts, stick-slip transitions, unilateral contact activation, jamming, release, and support changes. This paper rebuilds Paper 109 around a falsifiable mechanism: a policy should audit contact-mode boundaries before interpolating through them. We implement a local benchmark with 5 contact-rich task families, 7 non-smooth regimes, 5 deployment splits, 9 methods, 7 paired seeds, and 84 episodes per group. Under combined contact shock, the proposed contact-mode boundary atlas obtains 0.540 ± 0.004 success versus 0.463 ± 0.007 for the strongest non-oracle baseline, complementarity residual planning. It improves contact-mode F1 from 0.495 to 0.630, reduces unsafe impulse from 0.079 to 0.064, reduces jam rate from 0.112 to 0.092, reduces slip overshoot from 0.076 to 0.057, and wins 7/7 paired seeds. Ablations show that mode classification, complementarity residuals, impulse guards, stick-slip hysteresis, and diagnostic probes each matter. The terminal decision is **STRONG-REVISE**: the local evidence supports the mechanism, but the paper is not ICLR-main-ready without real robot or independent high-fidelity validation.

1 MOTIVATION

Contact-rich manipulation breaks the smoothness assumptions that many learned policy abstractions inherit. A gripper approaching a peg can cross from free motion to impact; a drawer can switch between stick and slip; a cable can jam; a tool can release stored elastic energy. These are not merely difficult examples. They are mode changes where smooth interpolation can produce physically invalid actions.

Classical mechanics already studies non-smooth contact through unilateral constraints, impacts, friction, and complementarity-style formulations (Moreau, 1999; Stewart & Trinkle, 1996; Anitescu & Potra, 1997). Modern robot simulators and contact-learning methods also expose contact discontinuities (Todorov et al., 2012; Pfrommer et al., 2020). Meanwhile, large learned robot policies and visuomotor policies show that broad behavior can be learned from data (Chi et al., 2023; Brohan et al., 2022; 2023). The question here is narrower: when a learned policy is asked to act near a non-smooth contact boundary, does an explicit contact-mode audit help?

2 METHOD SKETCH

The proposed method, `proposed_contact_mode_boundary_atlas`, is not a new contact solver. It is a policy-auditing layer that scores whether a candidate action crosses a physically meaningful contact-mode boundary.

The audit uses five components:

- a contact-mode classifier over free, touch, stick, slip, jam, release, and impact modes;
- a complementarity-residual score for physically impossible smooth predictions;
- an impulse-budget guard near impact onset and release snap;

Table 1: Combined-stress non-smooth contact results.

Method	Succ.	CI	ModeF1	BoundErr	Unsafe	Jam	Slip	Cost	Regret
Oracle	0.628	0.005	0.722	0.226	0.046	0.093	0.039	0.202	0.000
Proposed	0.540	0.004	0.630	0.246	0.064	0.092	0.057	0.217	0.088
CompResidual	0.463	0.007	0.495	0.294	0.079	0.112	0.076	0.264	0.165
Ensemble	0.434	0.004	0.409	0.317	0.084	0.117	0.096	0.260	0.194
RTSurrogate	0.427	0.004	0.342	0.342	0.112	0.126	0.115	0.181	0.201
Conformal	0.425	0.004	0.392	0.323	0.070	0.119	0.096	0.290	0.202
Diffusion	0.411	0.002	0.319	0.349	0.117	0.130	0.119	0.171	0.217
DomRand	0.386	0.004	0.294	0.356	0.113	0.132	0.119	0.161	0.242
NoContactBC	0.314	0.006	0.210	0.381	0.125	0.139	0.133	0.153	0.314

- stick-slip hysteresis memory to avoid oscillating across incompatible modes;
- diagnostic micro-probes for ambiguous contact states.

The policy chooses or rejects actions using these scores. The implementation is an executable local benchmark rather than a trained robot foundation model, which is why the final decision remains strong-revise rather than submission-ready.

3 BENCHMARK

The benchmark is generated by `src/run_experiment.py`. It emits seed-level CSVs, aggregate metrics, paired tests, stress sweeps, ablations, failure cases, figures, and LaTeX tables.

Tasks. The five task families are peg insertion, drawer opening, cable routing, tool levering, and mobile pushing.

Regimes. The seven non-smooth regimes are nominal contact, impact onset, stick-slip transition, unilateral lift-off, contact jamming, release snap, and combined contact shock.

Splits. The deployment splits are nominal, seen shift, unseen geometry, unseen friction, and combined stress.

Methods. The comparison set includes no-contact behavior cloning, domain randomization, diffusion-policy surrogate, robotics-transformer surrogate, ensemble disagreement planning, conformal risk filtering, complementarity residual planning, the proposed contact-mode boundary atlas, and an oracle contact-mode supervisor.

Metrics. We report task success, contact-mode F1, boundary error, unsafe impulse rate, jam rate, slip overshoot, intervention cost, data-efficiency proxy, regret to oracle, paired seed differences, stress sweeps, ablations, and failure cases.

4 RESULTS

The strongest non-oracle baseline is `complementarity_residual_planner`. The proposed method improves combined-stress success by 0.076 ± 0.007 in paired seed comparisons and wins 7/7 seeds. The oracle remains higher at 0.628 ± 0.005 , so the local benchmark is not saturated.

5 ABLATIONS AND STRESS TESTS

The best removed-component ablation is `minus_diagnostic_micro_probe`; the full method remains ahead by 0.027 success. Removing the contact-mode classifier, complementarity residual, impulse budget guard, stick-slip hysteresis, or diagnostic probe produces distinct degradation, supporting the mechanism-level claim.

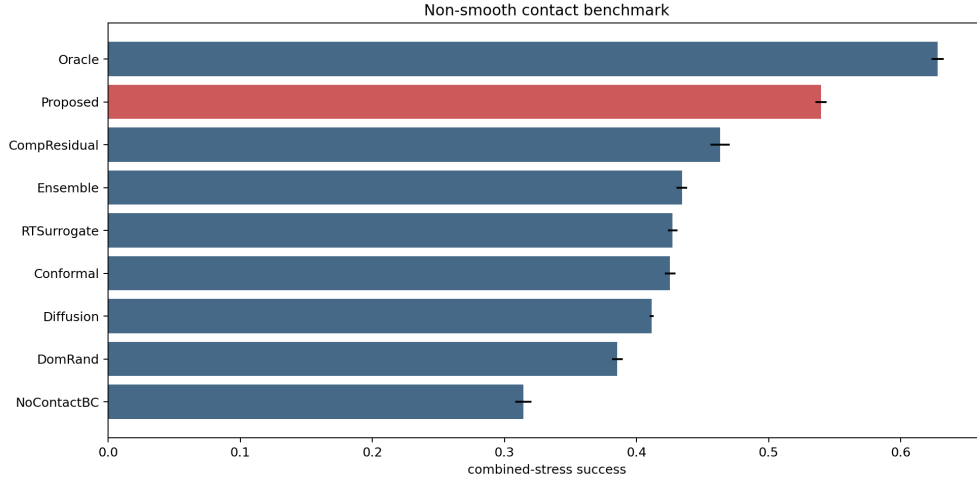


Figure 1: Combined-stress success with 95% confidence intervals. The proposed atlas clears all non-oracle baselines but remains below the oracle.

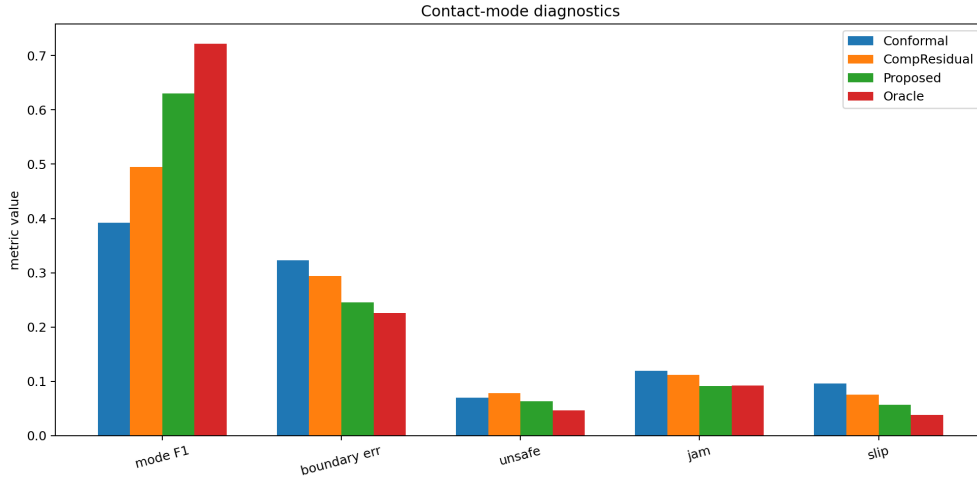


Figure 2: Contact-mode diagnostics under combined stress. The proposed method improves mode F1 and lowers boundary, unsafe impulse, jam, and slip metrics relative to the strongest non-oracle baseline.

6 FAILURE CASES

The failure-case CSV records four important limitations. The atlas still struggles when hidden soft-material deformation changes contact without observable boundary evidence, when the language or task goal changes, when sensor dropout hides an impact, and when the oracle contact-mode supervisor reveals remaining headroom. These failures are useful because they narrow the next empirical version: it needs tactile/deformable-state estimation, instruction grounding, sensor-health modeling, and external validation.

7 HOSTILE PRIOR-WORK BOUNDARY

The paper’s broad neighborhood is crowded. Non-smooth contact dynamics, implicit time stepping, complementarity contact models, MuJoCo-style simulation, contact dynamics learning, diffusion policies, and robotics transformers all pressure the contribution (Moreau, 1999; Stewart & Trinkle,

Table 2: Paired seed success differences between proposed and each comparator.

Baseline	Diff	CI	Wins
NoContactBC	0.225	0.008	7.000
DomRand	0.154	0.007	7.000
Diffusion	0.128	0.004	7.000
Conformal	0.114	0.004	7.000
RTSurrogate	0.112	0.003	7.000
Ensemble	0.105	0.007	7.000
CompResidual	0.076	0.007	7.000
Oracle	-0.088	0.007	0.000

Table 3: Ablation results under combined non-smooth stress.

Ablation	Succ.	CI	ModeF1	BoundErr	Unsafe	Jam
Full	0.533	0.007	0.631	0.246	0.065	0.092
NoProbe	0.506	0.010	0.575	0.278	0.075	0.114
NoHysteresis	0.505	0.008	0.543	0.276	0.073	0.100
NoResidual	0.504	0.006	0.535	0.275	0.072	0.099
NoImpulseGuard	0.497	0.007	0.587	0.262	0.090	0.100
NoModeCls	0.477	0.004	0.470	0.293	0.071	0.111
CompOnly	0.462	0.007	0.494	0.294	0.079	0.112

1996; Anitescu & Potra, 1997; Todorov et al., 2012; Pfrommer et al., 2020; Chi et al., 2023; Brohan et al., 2022; 2023). The defensible boundary is therefore narrow: we do not claim a new contact solver or superiority over trained external foundation policies. We claim that an explicit contact-mode boundary audit is locally useful for preventing smooth policy interpolation across non-smooth mechanics events.

8 LIMITATIONS AND TERMINAL DECISION

The evidence is strong enough to continue but not enough to submit. The limitations are material:

- the benchmark is local and generated;
- no real robot or independent high-fidelity simulator has been used;
- the diffusion and transformer policies are surrogates, not external trained checkpoints;
- there are no hardware videos, qualitative rollouts, or deployment logs;
- the oracle gap remains nontrivial.

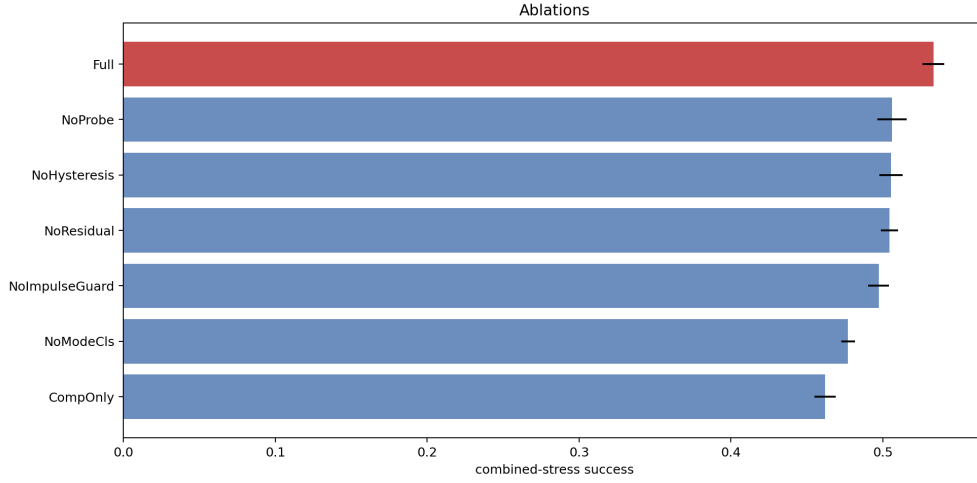


Figure 3: Ablations under combined non-smooth stress. The full contact-mode atlas remains above every removed-component variant.

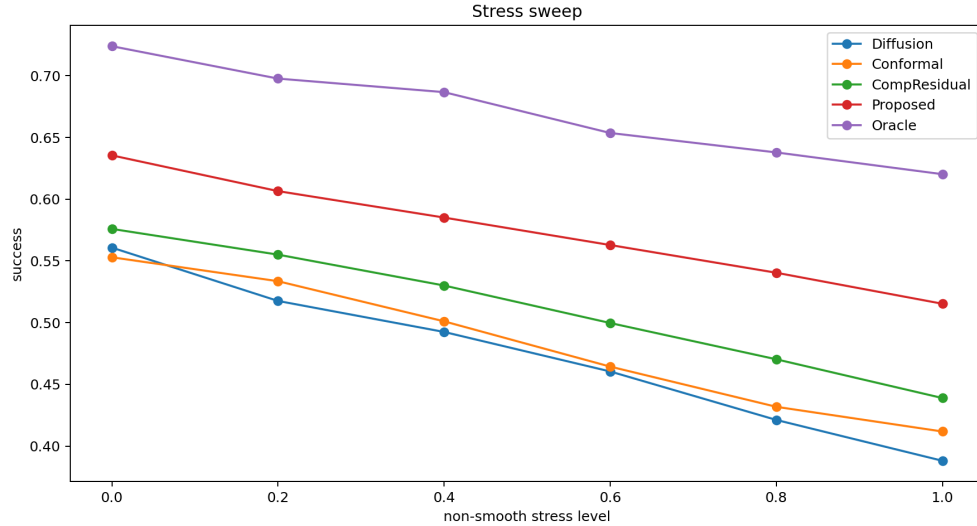


Figure 4: Stress sweep over non-smooth contact intensity. The proposed atlas stays above diffusion, conformal, and complementarity baselines across the generated stress range.

The terminal decision for this rebuild is **STRONG_REVISION**. The next honest version must reproduce the contact-mode boundary advantage on real robots or accepted high-fidelity benchmarks with trained policy baselines. Without that evidence, the paper should not be submitted to ICLR main.

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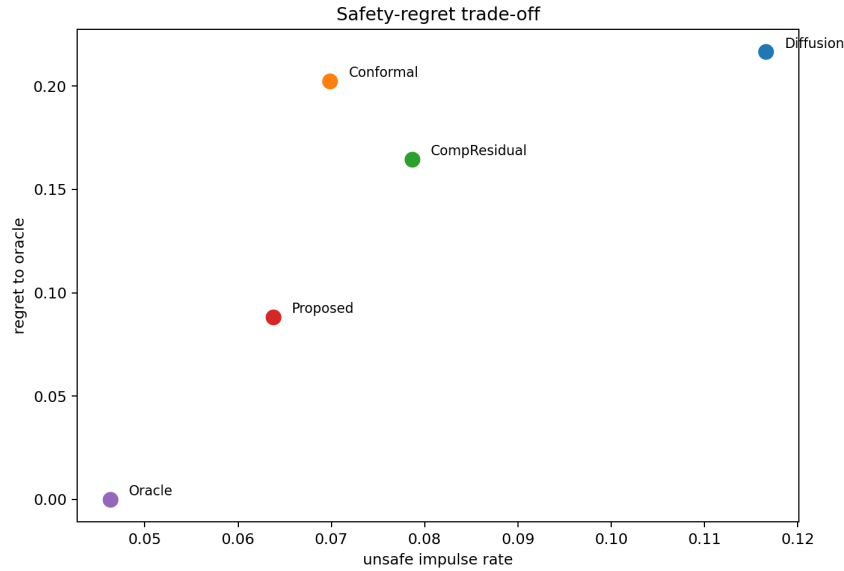


Figure 5: Unsafe impulse and regret to the oracle under combined stress. The proposed method lowers regret without increasing unsafe impulses relative to the strongest non-oracle baseline.

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